

Signal Processing

Exercise Sheet 9 – IIR Filter

SoSe 2026

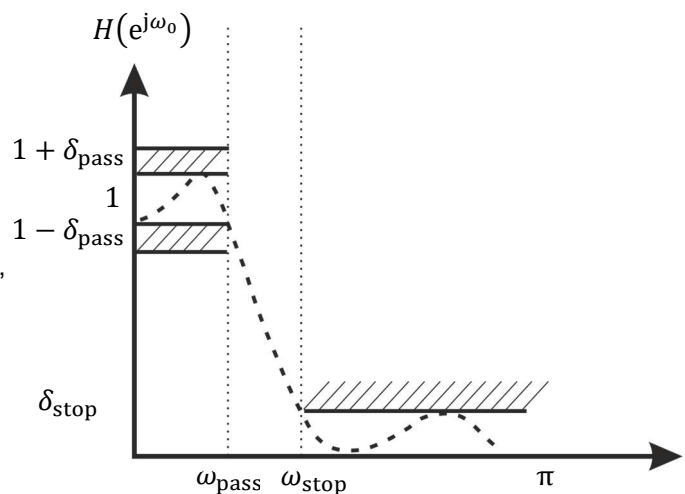
Digital Filter Design - Basics



There is a rich collection of available filter designs for analog filters which are based on already well known and efficient methods. It is for this reason, that procedures for digital filter design are directed at transforming analog filters to digital filters by taking advantage of well-established design procedures. However, as with analog filters, digital filters will deviate from the ideal response as well.

An important property of linear shift invariant systems (LSI) is that a complex exponential input is carried over to a complex exponential output of the same frequency multiplied by the system function $H(e^{j\omega_0})e^{j\omega_0 n}$. This allows us to consider a signal to actually consist of the sum of two (or more) separate signals, one (few) we would like to suppress and one (others) we would like to keep. Due to this separation, a proper choice of the system function $H(e^{j\omega_0})$ allows us to reject frequency components of no interest and keeping the desired ones at the same time.

$$e^{j\omega_0 n} \longrightarrow H(e^{j\omega_0})e^{j\omega_0 n}$$



Thus, we can think of designing a filter as the implementation of a linear shift invariant system with a particular frequency response characteristic.

There are two common ways to design such filters. The first one is referred to as the *continuous time to discrete time method*, the second one is referred to as the *algorithmic (computer aided) method*. We will make use of the first one in this exercise lesson but will keep in mind that the latter is the to-go-to procedure nowadays.

Since it is convenient and we restricted ourselves in the lectures to system functions whose input-output characteristics can be expressed with the help of a linear constant coefficient difference equation, an ideal filter cannot be implemented. Thus, we will need to allow for some deviation and approximate the ideal frequency characteristics.

To sum up, we would like to map a continuous time system function in the Laplace plane to a digital system function (z plane). We desire to do so in such a way that a good filter in the s plane corresponds to a good filter in the z plane. This requirement results in two restrictions we need to keep in mind if we evaluate different mapping procedures:

1. $j\Omega$ – axis maps to the unit circle.
2. If the system function $H(s)$ is stable, then the system function $H(z)$ should be stable as well.

There are three methods that we will discuss in this exercise lesson.

1. Replacing the Derivatives by Differences

This procedure is the most intuitive but has several downsides in terms of the two guidelines we set down.

In order to derive this procedure, we will have a look at the derivative in the s plane

$$\mathcal{L}\left[\frac{dy(t)}{dt}\right] = s \cdot Y(s) \quad (1)$$

and its counterpart the difference in the z plane

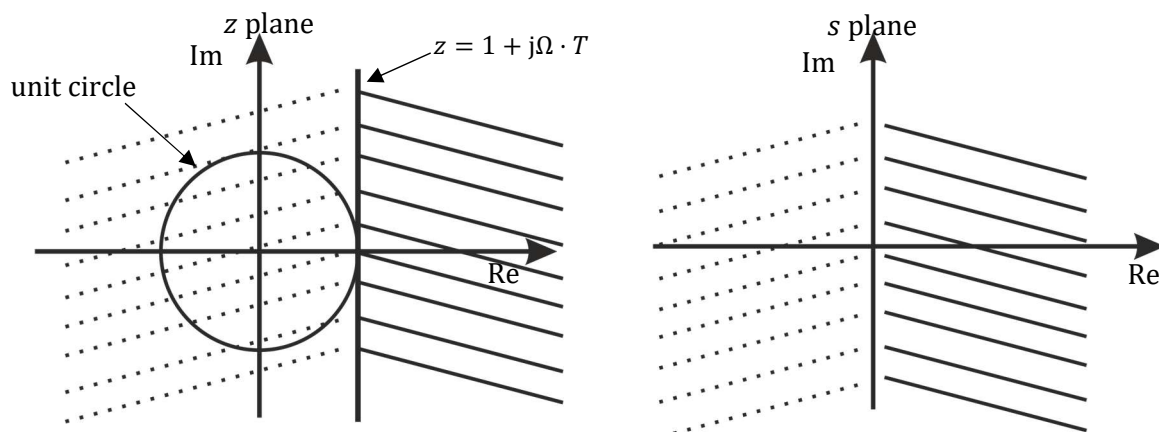
$$\mathbb{Z}\left[\frac{y(n+1) - y(n)}{T}\right] = \frac{z-1}{T} Y(z). \quad (2)$$

Comparing equation (1) and (2) indicates that the mapping corresponding to the replacement of differentials by differences reads

$$s = \frac{z-1}{T} \quad (3)$$

or

$$z = 1 + s \cdot T. \quad (4)$$



Disadvantages:

- A stable filter in the s plane does not guarantee a stable filter in the z plane.
- The $j\Omega$ axis does not map to the unit circle. Thus, only at extremely high oversampling rates the filters, which showed a good system function in the s plane, will show similar results in the z plane.

2. Impulse Invariance

The idea of the impulse invariance method is to obtain the discrete-time sample response by sampling the analog impulse response.

The digital frequency response is a scaled replica of the analog frequency response. Unfortunately, this mapping method leads to aliasing in the digital frequency response.

We will no further discuss this procedure in detail but will keep in mind that this method is still commonly used, although the designer needs to carefully decide on the frequency components where aliasing does not affect the system concept/application.

3. Bilinear Transform

The procedure of *bilinear transform* allows for avoiding aliasing by mapping the $j\Omega$ axis to the unit circle. In addition, stable analog filters in the s plane result in stable filters in the z plane. However, the procedure introduces a distortion in the frequency axis. Depending on the application, this distortion can be tolerated as long as the filter is piece-wise constant at the frequency components of interest. This technique is derived by a continuous time differential equation which is integrated to receive an integral equation. The integral is then approximated by using the trapezoidal rule.

At first, we will start with a continuous time differential equation. In integrating this equation, we obtain

$$y(t) = \int_0^t x(\tau) d\tau, \quad (5)$$

provided we consider a causal system. This corresponds to

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s} \quad (6)$$

in the Laplace domain. If we approximated the integral with the help of the trapezoidal rule

$$y[n] = y[n-1] + \frac{x[n] + x[n-1]}{2} \cdot T, \quad (7)$$

our system function results in

$$H(z) = \frac{Y(z)}{X(z)} = \frac{T}{2} \cdot \frac{z+1}{z-1}. \quad (8)$$

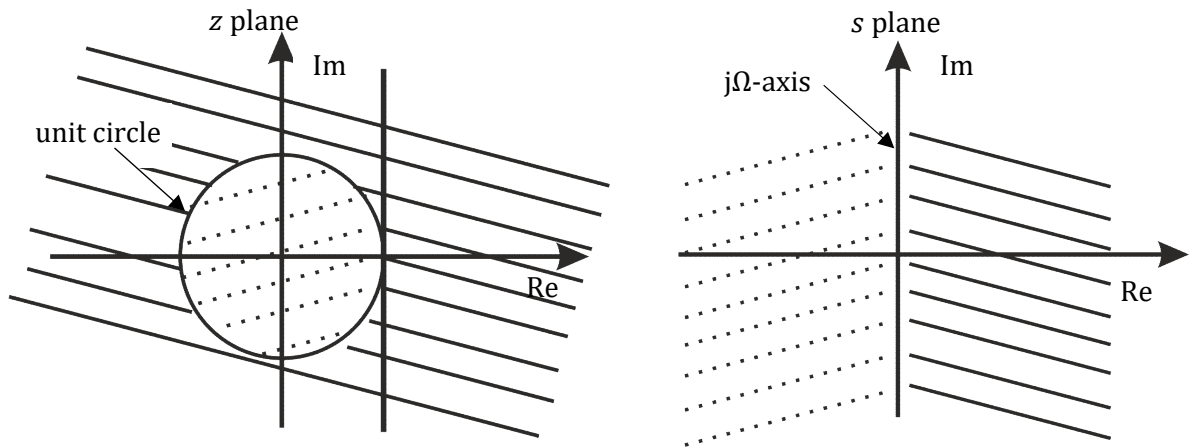
The comparison of equation (8) and equation (6) yields to our mapping rule

$$s = \frac{2}{T} \cdot \frac{z-1}{z+1} = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}, \quad (9)$$

and

$$z = \frac{1 + \frac{sT}{2}}{1 - \frac{sT}{2}}, \quad (10)$$

which is referred to as *the bilinear transform*. This relationship between the s plane and the z plane results in mapping the left side of the s plane to the inner part of the unit circle and the mapping of the right side of the s plane to the outer part of the unit circle. This implies that a causal and stable filter design in the s plane results in a stable filter design in the z plane, too.



One additional important feature of the bilinear transform is its mapping of the $j\Omega$ axis to the unit circle. It is for this reason that we do not have to account for aliasing as we had to do in case of the impulse invariance method. In fact, the entire $j\Omega$ axis ends up exactly **once** on the unit circle. However, since the $j\Omega$ axis is infinitely long but the unit circle's circumference is finite, this method introduces a distortion.

Since we would only like to evaluate the relationship between the $j\Omega$ axis and the unit circle, we substitute

$$z = e^{j\omega}, \quad (11)$$

which will allow us to assess to what extent the frequency characteristics are preserved. Inserting equation (11) in equation (9) yields

$$s = \frac{2}{T} \cdot \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}}. \quad (12)$$

It can be shown that equation (12) can be manipulated in such a way that it reads

$$s = \frac{2}{T} \cdot \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = \frac{2}{T} j \tan \frac{\omega}{2}. \quad (13)$$

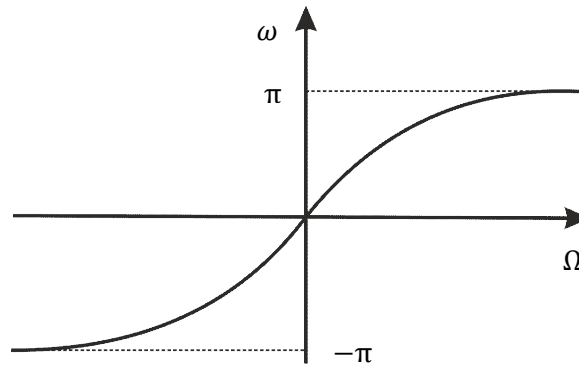
Since we only focus on the $j\Omega$ axis, we can replace s by $j\Omega$. Thus, equation (13) takes the form

$$j\Omega = \frac{2}{T} j \tan \frac{\omega}{2}, \quad (14)$$

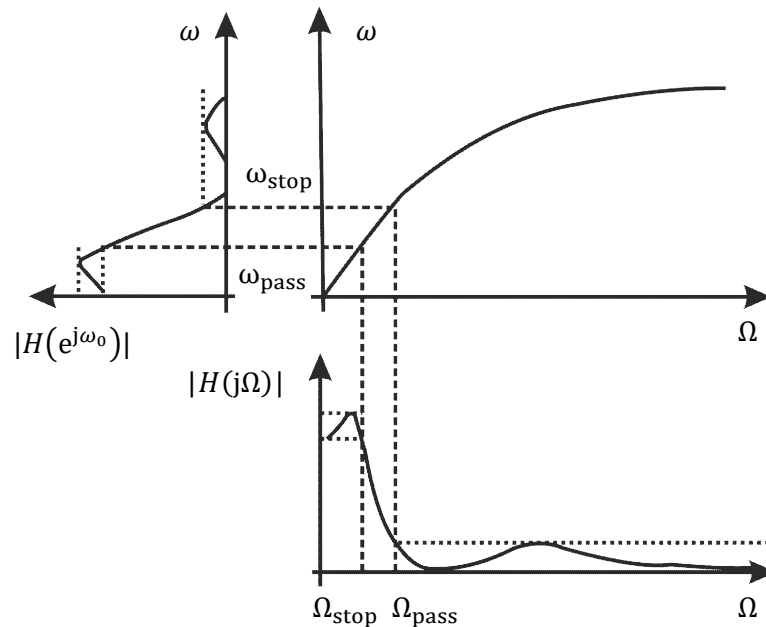
which can be simplified to

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}. \quad (15)$$

The following figure shows the relationship of Ω and ω .



If we have a closer look at the distortion of our pre-defined filter characteristics in the s plane mapped to the z plane, the frequency warping affects Ω_{pass} , Ω_{stop} and also its characteristics in the transition band. However, the maximum ripple we allow for both the passband and the stopband remains constant. Sometimes, the continuous-time cutoff frequencies are prewarped to achieve the desired discrete-time cutoff frequencies. Alternatively, if our application allows us to restrict ourselves to frequencies much lower than our sampling frequency, we could restrict our filter design to the almost linear relationship between Ω and ω .



Question 1

A digital filter is to be designed using the bilinear transform. Its specifications read

- $0.99 < |H(e^{j\omega})| \leq 1$, for $0 \leq \omega < \frac{\pi}{8}$
- $|H(e^{j\omega})| \leq 0.001$, for $\frac{\pi}{6} \leq \omega < \pi$
- a) Determine what kind of filter is described.
- b) Determine the specification on an analog filter which will yield the desired digital filter when the bilinear transform is applied with $T = 1$.

- a) Lowpass filter
- b) The relationship between Ω and ω reads

$$\Omega = 2 \cdot \tan \frac{\omega}{2}$$

This means we need

- $0.99 < |H(e^{j\omega})| \leq 1$, for $0 \leq \Omega < 2 \tan \frac{\pi}{16}$
- $|H(e^{j\omega})| \leq 0.001$, for $2 \tan \frac{\pi}{12} < \Omega$

Question 2

An analog highpass filter can be obtained from an analog lowpass filter by replacing s by $\frac{1}{s}$ in the system function. This means if $G_{La}(s)$ is the system function of a lowpass filter, then $H(s)$ is the system function of a highpass filter if

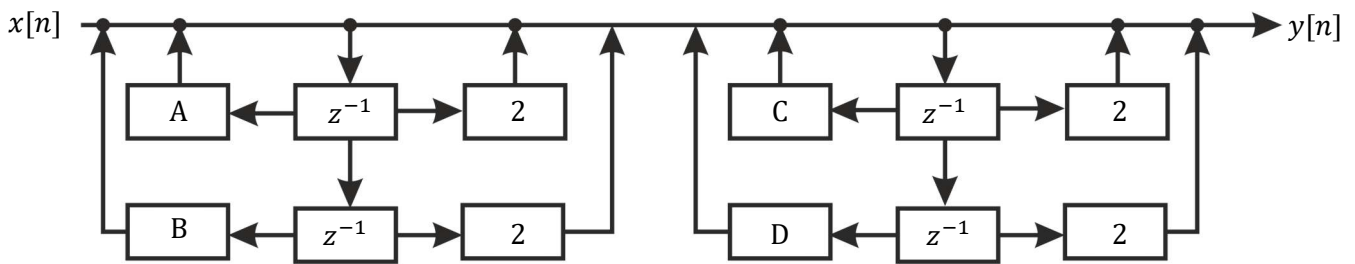
$$G_{La}\left(\frac{1}{s}\right) = H_{Ha}(s)$$

holds.

Since we would like to use the bilinear transform to map our analog filter to the z plane, we will make use of the mapping rule

$$s = \frac{z - 1}{z + 1},$$

even though the frequency response is distorted. The network in the following figure depicts a lowpass filter with a cutoff frequency $\omega = \frac{\pi}{2}$.



Determine how to modify the coefficients A , B , C and D in the network to obtain a highpass filter with a cutoff frequency $\omega_{ch} = \frac{\pi}{2}$. Note that A , B , C and D are real numbers and you need to determine a manipulation, not a value.

If we use the bilinear transform and the system function of the lowpass filter reads

$$G_{Lz}(z) = G_{La}\left(\frac{z - 1}{z + 1}\right), \quad (16)$$

the digital highpass filter yields

$$H(z)_{Hz} = H_{Ha}\left(\frac{z - 1}{z + 1}\right) = G_{La}\left(\frac{z + 1}{z - 1}\right) \quad (17)$$

If we compare $H(z)_{Hz}$ and $G_{Lz}(z)$ we can conclude that

$$H(z)_{Hz} = G_{Lz}(-z). \quad (18)$$

This implies that multiplying the output of each delay z^{-1} by -1 will result in a highpass filter if our choice of design is the bilinear transform.

In case we would like to determine to what extent the cutoff frequency ω_{CH} of our highpass filter relates to our cutoff frequency ω_{CL} of our lowpass filter, we replace z by

$$z = e^{j\omega}$$

and make use of the unit circle to represent -1 by

$$-1 = e^{j\pi}.$$

As a result, equation (18) takes the form

$$H(e^{j\omega})_{\text{Hz}} = G_{\text{Lz}}(e^{j\pi}e^{j\omega}) = G_{\text{Lz}}(e^{j(\pi+\omega)}). \quad (19)$$

Equation (19) shows that the frequency response is shifted by π and thus the cutoff frequency ω_{CH} of the highpass filter reads

$$\omega_{\text{CH}} = \pi - \omega_{\text{CL}} = \pi - \frac{\pi}{2} = \frac{\pi}{2}.$$

In fact, we could keep the coefficients as they are and only multiply each delay with -1. However, since the question clearly states we should modify A, B, C and D, we will make use of the relationship

$$(-z)^{-2} = z^{-2}.$$

Therefore, we will only change the sign of the coefficient whose input passes only an odd number of delays and keep all others as they are. Therefore, the two coefficients 2 and A, C will be multiplied by -1.

Question 3

We would like to design a discrete lowpass butterworth filter in the z plane which meets following criteria

- Maximum passband attenuation: 3 dB
- Minimum stopband attenuation: 20 dB
- Passband edge: 6 Hz
- Stopband edge: 16 Hz
- Sampling frequency: 48 Hz

The equation for the Butterworth filter reads either

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

or

$$|H_c(s)|^2 = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}}.$$

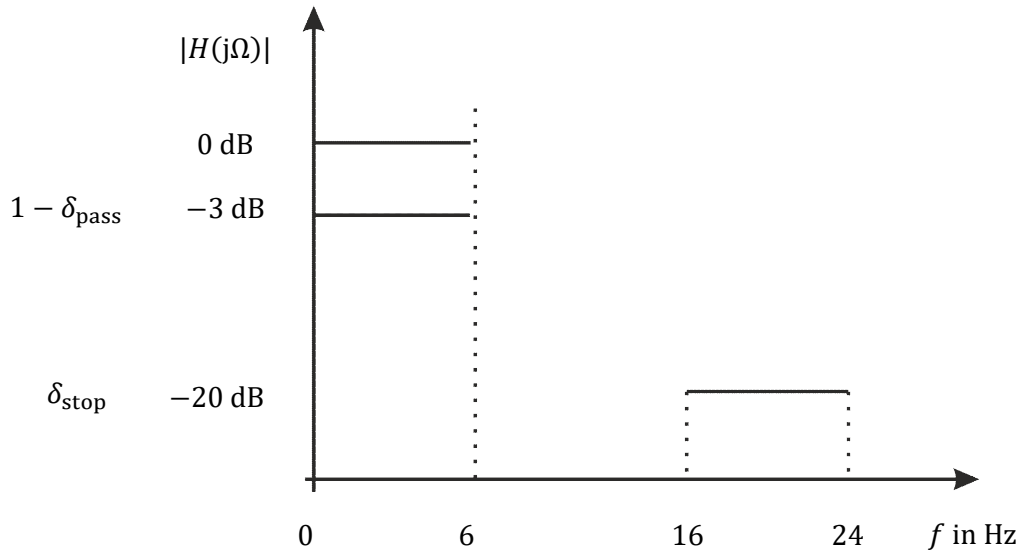
Thus, N is given by

$$N = \frac{1}{2} \frac{\log\left(\frac{1}{|H_c(j\Omega_{\text{pass}})|^2} - 1\right) - \log\left(\frac{1}{|H_c(j\Omega_{\text{stop}})|^2} - 1\right)}{\log \Omega_{\text{pass}} - \log \Omega_{\text{stop}}}.$$

In addition, we would like to make sure that the maximum passband attenuation is met exactly.

- a) Sketch the tolerance scheme of the filter.
- b) Decide on whether δ_{pass} or δ_{stop} imposes more restrictions on our design.
- c) Determine ω_{pass} , ω_{stop} , Ω_{pass} , Ω_{stop} and Ω_c and calculate the minimum required N . Assume $T = 1$.
- d) Determine a digital filter system function $H(z)$ by using the bilinear transform with $T = 1$. Hint: Determine $|H_c(s)|^2$ and $H_c(s)$ first. In order to do so, try to obtain the poles of $|H_c(s)|^2$.

a) Sketch the tolerance scheme of the filter.



b) Decide on whether δ_{pass} or δ_{stop} imposes more restrictions on our design.

$$20 \log_{10}(1 - \delta_{\text{pass}}) = -3 \text{ dB} \quad 1 - \delta_{\text{pass}} = 10^{-\frac{3}{20}} = 0.7 \rightarrow \delta_{\text{pass}} = 0.3$$

$$20 \log_{10} \delta_{\text{stop}} = -20 \text{ dB} \rightarrow \delta_{\text{stop}} = 10^{-\frac{20}{20}} = 0.1$$

$\delta_{\text{pass}} > \delta_{\text{stop}}$ means that the requirement on the stopband is higher than on the passband. If we can design a filter that fulfills the stopband specifications, the passband specifications will be fulfilled automatically.

Watch out! We cannot compare δ_{pass} and δ_{stop} directly on the dB specification in this case as we need to account for $(1 - \delta_{\text{pass}})$!

c) Determine ω_{pass} , ω_{stop} , Ω_{pass} , Ω_{stop} and Ω_c and calculate the minimum order N .

$$\omega_{\text{pass}} = \frac{2\pi \cdot 6}{48} = \frac{\pi}{4}$$

$$\omega_{\text{stop}} = \frac{2\pi \cdot 16}{48} = \frac{2\pi}{3}$$

$$\Omega_{\text{pass}} = 2 \cdot \tan\left(\frac{\omega_{\text{pass}}}{2}\right) = 0.83$$

$$\Omega_{\text{stop}} = 2 \cdot \tan\left(\frac{\omega_{\text{stop}}}{2}\right) = 3.46$$

Calculate the minimum required N .

Since our filter is designed to yield $|H_c(j\Omega_{\text{pass}})| > -3\text{dB} \approx 0.7$ and $|H_c(j\Omega_s)| < -20\text{dB} = 0.1$, we obtain

$$N = \frac{1}{2} \frac{\log\left(\frac{1}{|0.7|^2} - 1\right) - \log\left(\frac{1}{|0.1|^2} - 1\right)}{\log 0.82 - \log 3.46} = 1.58 \rightarrow N = 2$$

We choose N to be 2 since one always needs to round up to make sure all specifications are met.

The equation of the Butterworth filter can be solved for Ω_c which yields

$$\Omega_c = \frac{\Omega}{\sqrt[4]{\frac{1}{|H_c|^2} - 1}}$$

Since $0.7 \approx -3 \text{ dB} = H_c$ and we would like to meet the passband attenuation exactly, i.e. $\Omega = \Omega_{\text{pass}}$, we obtain

$$\Omega_c \approx 0.82.$$

- d) Determine a digital filter system function $H(z)$ by using the bilinear transform with $T = 1$. Hint: Determine $|H_c(s)|^2$ and $H_c(s)$ first. In order to do so, try to obtain the poles of $|H_c(s)|^2$.

In order to design our filter, the first step involves solving for the poles by setting the denominator $1 + \left(\frac{s}{j\Omega_c}\right)^{2N}$ to zero. This will allow us to solve the equation for the Butterworth filter more easily.

Therefore, we obtain

$$s = j \Omega_c \sqrt[2N]{-1}.$$

Taking the relationships $-1 = e^{j(\pi+2\pi k)}$ and $j = e^{j\frac{\pi}{2}}$ into account, our equation takes the form

$$s = \Omega_c e^{j\frac{3\pi+2\pi k}{4}}, \text{ for } k = 0, \dots, 3.$$

By setting $|H_c(s=0)|^2 = 1$ we obtain

$$|H_c(s)|^2 = \frac{\Omega_c^2 \cdot \Omega_c^2}{(s - \Omega_c e^{j\pi/4})(s - \Omega_c e^{j3\pi/4})(s - \Omega_c e^{j5\pi/4})(s - \Omega_c e^{j7\pi/4})}$$

If we consider that $|H_c|^2$ consist of two functions multiplied with one another

$$|H_c(s)|^2 \triangleq G(s)^2 = H_{c-1}(s) \cdot H_{c-1}(-s),$$

one that is stable and one that is not stable, we will only opt for the stable part which reads

$$H_{c-1}(s) = \frac{\Omega_c^2}{(s - \Omega_c e^{j3\pi/4})(s - \Omega_c e^{-j3\pi/4})} = \frac{0.687}{s^2 + 1.17s + 0.687}.$$

In using our mapping rule of the bilinear transform, we receive our final system filter function

$$H(z) = \frac{0.0977 + 0.195z^{-1} + 0.0977z^{-2}}{1 - 0.943z^{-1} + 0.334z^{-2}}.$$